

QMST5336: Analytics

Lecture 3: Random Variables

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Random Variables

Definitions:

- **Random variable:** is a numeric quantity whose value depends on the outcome of a random event
 - We use a capital letter, like X , to denote a random variable
 - The values of a random variable are denoted with a lowercase letter, in this case x
 - For example, $P(X = x)$

- **Probability distribution:** assigns probabilities to the values that the random variable can take

Note: In this lecture we are dealing with *all possible outcomes*.

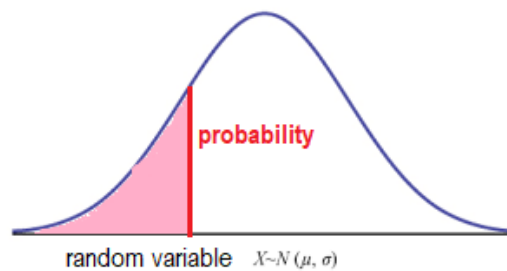
Random Variables

Example 1: A basketball player shoots three free throws.

- Let X be a random variable representing the number of baskets scored
- X has the probability distribution:

Value of X	$X = 0$	$X = 1$	$X = 2$	$X = 3$
Probability	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Example 2: *Normal Distribution*



Random Variables

There are two types of random variables:

- **Discrete:** the variables outcomes that can be given in an ordered list (you can count them)
 - Example: The outcome of a throwing a dice $\{1, 2, 3, 4, 5, 6\}$
- **Continuous:** the variables outcomes can take any value from a range of values
 - Example: The height of students in this class follows a **normal distribution** with mean 1.7 meters and standard deviation 0.1 meters

Discrete: Expected value & Variability

Let variable X take values from the set $\{x_1, \dots, x_N\}$

- The **expected value** μ_X of a random variable X is a **weighted average of all possible values of X**

$$\mu_X = \sum_{i=1}^N x_i \cdot P(X = x_i)$$

- The **variability** σ_X^2 of a random variable is a **weighted average of the squared deviations** $(X - \mu_X)^2$ of the variable X from its mean μ_X

$$\sigma_X^2 = \sum_{i=1}^N (x_i - \mu_X)^2 \cdot P(X = x_i)$$

Discrete: Expected and Variability

Example: A basketball player shoots three free throws. $x_1 = 0$, $x_2 = 1$, $x_3 = 2$, $x_4 = 3$

Value of X	$X = 0$	$X = 1$	$X = 2$	$X = 3$
Probability	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$$\begin{aligned}\mu_X &= \sum_{i=1}^N x_i \cdot P(X = x_i) \\ &= 0 \cdot P(X = 0) + 1 \cdot P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) \\ &= 0 \left(\frac{1}{8}\right) + 1 \left(\frac{3}{8}\right) + 2 \left(\frac{3}{8}\right) + 3 \left(\frac{1}{8}\right) = 1.5\end{aligned}$$

$$\begin{aligned}\sigma_X^2 &= \sum_{i=1}^N (x_i - \mu_X)^2 \cdot P(X = x_i) \\ &= (0 - 1.5)^2 \cdot P(X = 0) + (1 - 1.5)^2 \cdot P(X = 1) \\ &\quad + (2 - 1.5)^2 \cdot P(X = 2) + (3 - 1.5)^2 \cdot P(X = 3) \\ &= 2.25 \left(\frac{1}{8}\right) + 0.25 \left(\frac{3}{8}\right) + 0.25 \left(\frac{3}{8}\right) + 2.25 \left(\frac{1}{8}\right) = 0.75\end{aligned}$$

Addition of Expected values and Variability

Addition of Expected values: If X and Y are random variables, and α and β are weights.

- Then expected values of $\alpha X + \beta Y$ is equal to $\alpha\mu_X + \beta\mu_Y$
- Then expected values of $\alpha X - \beta Y$ is equal to $\alpha\mu_X - \beta\mu_Y$

Addition of Variability: If X and Y are random variable with correlation coefficient r , and α and β are weights.

- Then variability of $\alpha X + \beta Y$ is equal to $\alpha^2\sigma_X^2 + \beta^2\sigma_Y^2 + 2r(\alpha\sigma_X)(\beta\sigma_Y)$
- Then variance of $\alpha X - \beta Y$ is equal to $\alpha^2\sigma_X^2 + \beta^2\sigma_Y^2 - 2r(\alpha\sigma_X)(\beta\sigma_Y)$

Note:

- Correlation coefficient $-1 \leq r \leq 1$ represents the linear association of two random variables.
- If two random variables are independent, $r = 0$.

Addition of Expected values and Variability

Example: You invest your money as follows

- 20% in Stock X , and
- 80% in Stock Y

- Stock X has a \$5 mean annual return, and a standard deviation of \$2.9
- Stock Y has a \$13.2 mean annual return, and a std. deviation of \$17.6
- Correlation between the two stocks is $r = -0.11$

Question: How much money will your portfolio earn on average, and what is the variance of this portfolio?

Addition of Expected values and Variability

Our portfolio is the random variable $\alpha X + \beta Y = 0.2X + 0.8Y$, where $\alpha = 0.2$ and $\beta = 0.8$

- Expected value of $0.2X + 0.8Y$: We know $\mu_X = 5$ and $\mu_Y = 13.2$

$$\begin{aligned}\mu_{\alpha X + \beta Y} &= \alpha\mu_X + \beta\mu_Y \\ &= 0.2(5) + 0.8(13.2) = 11.56\end{aligned}$$

- Variability of $0.2X + 0.8Y$: We know $\sigma_X = 2.9$, $\sigma_Y = 17.6$, and $r = -0.11$

$$\begin{aligned}\sigma_{\alpha X + \beta Y}^2 &= \alpha^2\sigma_X^2 + \beta^2\sigma_Y^2 + 2r(\alpha\sigma_X)(\beta\sigma_Y) \\ &= 0.2^2(2.9)^2 + 0.8^2(17.6)^2 + 2(-0.11)(0.2 \cdot 2.9)(0.8 \cdot 17.6) \\ &= 196.786\end{aligned}$$

- Standard deviation of $0.2X + 0.8Y$: square root of variance of $0.2X + 0.8Y$

$$\sigma_{\alpha X + \beta Y} = \sqrt{\sigma_{\alpha X + \beta Y}^2} = \sqrt{196.786} = 14.03\%$$

Random Variable: Practice Question 1

The distribution function of a random variable X is shown below

Value of X	$X = 1$	$X = 2$	$X = 3$
Probability	0.2	0.6	0.2

The expected values of X is:

- A. 2
- B. 2.5
- C. 3
- D. 3.5

Answer: A. Using the definition

$$\begin{aligned}\mu_X &= \sum_{i=1}^N x_i \cdot P(X = x_i) \\ &= 1 \cdot P(X = 1) + 2 \cdot P(X = 2) + 3 \cdot P(X = 3) \\ &= 1(0.2) + 2(0.6) + 3(0.2) = 2\end{aligned}$$

Random Variable: Practice Question 2

The distribution function of a random variable X is shown below

Value of X	$X = 1$	$X = 2$	$X = 3$
Probability	0.2	0.6	0.2

The Variability of X is:

- A. 0.1
- B. 0.2
- C. 0.3
- D. 0.4

Answer: D. From the previous question we know that $\mu_X = 2$. Using the definition

$$\begin{aligned}\sigma_X^2 &= \sum_{i=1}^N (x_i - \mu_X)^2 \cdot P(X = x_i) \\ &= (1 - 2)^2 \cdot P(X = 1) + (2 - 2)^2 \cdot P(X = 2) + (3 - 2)^2 \cdot P(X = 3) \\ &= 1(0.2) + 0(0.6) + 1(0.2) = 0.4\end{aligned}$$

Random Variable: Practice Question 3

A random variable X has mean 5 and variance 2. Another random variable Y has mean 10 and variance 4.

The expected values of the random variable $X - Y$ is

- A. 15
- B. 5
- C. -5
- D. 4

Answer: C. We know that $\mu_X = 5$ and $\mu_Y = 10$. Using the definition

$$\mu_{\alpha X - \beta Y} = \alpha \mu_X - \beta \mu_Y$$

Since we are interested in the mean of $X - Y$, then $\alpha = 1$, $\beta = 1$

$$\begin{aligned}\mu_{X-Y} &= \mu_X - \mu_Y \\ &= 5 - 10 = -5\end{aligned}$$

Random Variable: Practice Question 4

A random variable X has mean 5 and variance 2. Another random variable Y has mean 10 and variance 4. Variables X and Y are independent. What is the variance of the random variable $X - Y$?

The variability of $X - Y$ is:

- A. 15
- B. 6
- C. 5
- D. 4

Answer: B. We know that $\sigma_X^2 = 2$ and $\sigma_Y^2 = 4$. Since there X and Y are uncorrelated, then $r = 0$. Using the definition

$$\sigma_{\alpha X - \beta Y}^2 = \alpha^2 \sigma_X^2 + \beta^2 \sigma_Y^2 - 2r(\alpha \sigma_X)(\beta \sigma_Y)$$

Since we are interested in $X - Y$, then $\alpha = 1$, $\beta = 1$

$$\begin{aligned}\sigma_{X-Y}^2 &= \sigma_X^2 + \sigma_Y^2 - 2r(\sigma_X)(\sigma_Y) \\ &= 2 + 4 = 6\end{aligned}$$

Random Variable: Practice Question 5

A casino game costs \$5 to play. If the first card you draw is red, then you get to draw a second card (without replacement). If the second card is the ace of clubs, you win \$500. If not, you don't win anything, i.e. lose your \$5. What is your expected profits/losses from playing this game? Remember: profit/loss = winnings - cost.

- A. A profit of 5 cents
- B. A loss of 10 cents
- C. A loss of 25 cents
- D. A loss of 30 cents

Event	Win	Profit: X	$P(X)$	$X \times P(X)$
Red, A♣	500	$500 - 5 = 495$	$\frac{26}{52} \times \frac{1}{51} = 0.0098$	$495 \times 0.0098 = 4.851$
Other	0	$0 - 5 = -5$	$1 - 0.0098 = 0.9902$	$-5 \times 0.9902 = -4.951$
				$E(X) = -0.1$

Fair game

A *fair* game is defined as a game that costs as much as its expected payout, i.e. expected profit is 0.

Do you think casino games in Vegas cost more or less than their expected payouts?

If those games cost less than their expected payouts, it would mean that the casinos would be losing money on average, and hence they wouldn't be able to pay for all this:

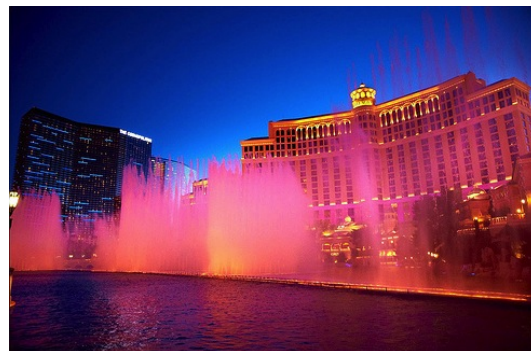


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